COVID-19 Sounds: A Large-Scale Audio Dataset for Digital Respiratory Screening

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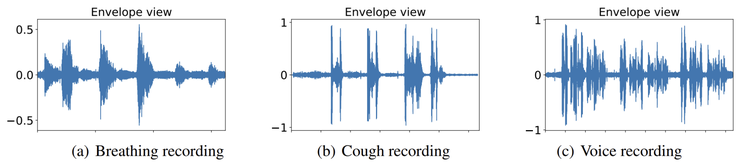
Audio signal based COVID-19 screening is affordable and promising, but current studies suffer from small datasets and unrealistic evaluations

Audio signals are widely recognised as powerful indicators of overall health status, and there has been increasing interest in leveraging sound for affordable COVID-19 screening through machine learning. However, there has also been scepticism regarding the initial efforts, due to perhaps the lack of reproducibility, large datasets and transparency which unfortunately is often an issue with machine learning for health. To facilitate the advancement and openness of audio-based machine learning for respiratory health, we release a dataset consisting of 53,449 audio samples (over 552 hours in total) crowd-sourced from 36,116 participants through our COVID-19 Sounds app.

Crowd-sourced Data Collection

COVID-19 Sounds App. The users are asked to input their symptoms along with medical history, as well as to record breathing, cough, and voice sounds every couple of days.

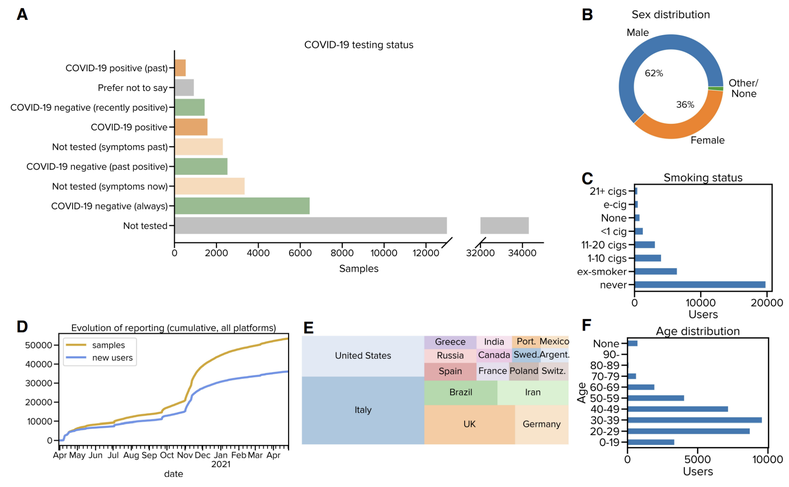


(a) Breathing recording (b) Cough recording (c) Voice recording

Audio recordings from a 30-year-old male, who was tested COVID-19 positive.

Highlights:

- Data collection: Compared to other publicly available audio datasets pertaining to COVID-19, our app is one of the earliest launched crowd-sourcing tools for COVID-19 sounds worldwide, and thus our collected data is large-scale with three sound types (breathing, cough, and voice).
- Meta-information: In addition to the COVID-19 status, most existing datasets also provide other metadata, such as age, gender, the presence of relevant symptoms, the presence of comorbidities (e.g., asthma, COPD, pulmonary fibrosis), and smoking status.



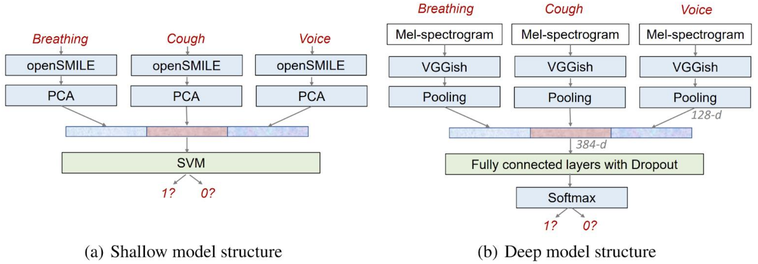
Dataset Overview. Our collected dataset consisting of 53,449 audio samples from 36,116 participants is released with this paper. It is one of the most comprehensive datasets in terms of global reach, number of languages, broad demographic range, span of health conditions, and number of audio modalities.

Data access: The study was approved by the ethics committee of the Department of Computer Science at the University of Cambridge, with ID #722. The data is sensitive, as voice can be deanonymised. It will be made available for academic research upon requests. Academic institutions will need to sign a Data Transfer Agreement with the University of Cambridge to obtain the data. We have this agreement in place; please contact covid-19-sounds@cl.cam.ac.uk to obtain it.

Benchmark Tasks

We examine two benchmark tasks and provide three baseline models including including both shallow and deep learning methods.

- **Respiratory symptom prediction**: distinguish between **healthy controls** and generic **respiratory abnormalities** such as dry cough, wet cough, fever, sore throat, shortness of breath, runny nose, headache, dizziness, and chest tightness
- **COVID-19 prediction**: distinguish between **tested-negative controls**, including symptomatic cases which could be caused by another infection and **tested-positive patients**, including asymptomatic cases



(a) Shallow model structure (b) Deep model structure

Baseline models. (a) We follow an established acoustic feature set which is then fed to SVM models. (b) For deep learning, we build upon a pre-trained neural network to extract generic audio features. We test training scenarios: a frozen encoder and a fine-tuned model tailored to the task at hand.

Website: https://www.covid-19-sounds.org/en/
Codes available at https://github.com/cam-mobsys/covid19-sounds-neurips

Experimental results

Evaluation

Modalities & baselines		Task 1 — 9,456 samples			Task 2 — 1,486 samples		
		ROC-AUC	Sensitivity	Specificity	ROC-AUC	Sensitivity	Specificity
Breathing	OpenSMILE+SVM	0.60(0.58-0.63)	0.47(0.44-0.50)	0.67(0.64-0.70)	0.56(0.50-0.61)	0.38(0.30-0.45)	0.70(0.64-0.77)
	Pre-trained VGGish	0.52(0.50-0.55)	0.07(0.05-0.08)	0.95(0.93-0.96)	0.59(0.52-0.65)	0.63(0.56-0.70)	0.49(0.41-0.56)
	Fine-tuned VGGish	0.65(0.63-0.67)	0.55(0.52-0.58)	0.66(0.63-0.69)	0.62(0.56-0.69)	0.64(0.56-0.71)	0.56(0.48-0.64)
Cough	OpenSMILE+SVM	0.70(0.67-0.72)	0.60(0.57-0.63)	0.65(0.62-0.68)	0.62(0.56-0.68)	0.56(0.48-0.63)	0.61(0.54-0.69)
	Pre-trained VGGish	0.66(0.63-0.68)	0.67(0.64-0.70)	0.53(0.50-0.56)	0.62(0.56-0.68)	0.69(0.61-0.76)	0.45(0.38-0.53)
	Fine-tuned VGGish	0.74(0.72-0.76)	0.70(0.67-0.73)	0.68(0.65-0.71)	0.66(0.59-0.71)	0.59(0.51-0.65)	0.66(0.59-0.73)
Voice	OpenSMILE+SVM	0.63(0.60-0.65)	0.56(0.53-0.59)	0.62(0.59-0.65)	0.52(0.45-0.58)	0.43(0.35-0.50)	0.62(0.54-0.69)
	Pre-trained VGGish	0.59(0.57-0.62)	0.56(0.53-0.59)	0.57(0.54-0.60)	0.61(0.54-0.67)	0.53(0.45-0.61)	0.66(0.59-0.74)
	Fine-tuned VGGish	0.69(0.66-0.71)	0.59(0.56-0.62)	0.67(0.64-0.70)	0.61(0.55-0.67)	0.57(0.49-0.65)	0.60(0.53-0.68)
Fusion	OpenSMILE+SVM	0.74(0.72-0.76)	0.65(0.62-0.68)	0.69(0.66-0.72)	0.64(0.58-0.70)	0.54(0.47-0.62)	0.67(0.60-0.75)
	Pre-trained VGGish	0.67(0.64-0.69)	0.61(0.58-0.64)	0.61(0.58-0.65)	0.64(0.58-0.70)	0.50(0.41-0.58)	0.63(0.56-0.70)
	Fine-tuned VGGish	0.75(0.73-0.77)	0.70(0.67-0.72)	0.70(0.67-0.72)	0.71(0.65-0.76)	0.65(0.57-0.72)	0.69(0.62-0.77)

Findings

- **Accuracy**: Audio shows great promise for these two tasks (AUC > 0.7)
- **Models:** Fine-tuned transfer learning-based deep model yielded the best performance when using all 3 audio modalities together.
- **Bottleneck**: A participant's symptom onset and other comorbidities represent a real challenge to audio-based COVID-19 screening.
- **Improvement:** The current performance may not satisfy the safety requirements for clinical diagnostics, which will be future work.

Potential Applications and Research Directions

- **Symptom based COVID-19 prediction:** In addition to audio, other auxiliary information we collected could itself be of great value. Symptoms along with demographics can be employed for real-time COVID-19 tracking.
- **Biometric user authentication**: Our dataset could help scale breathing-based user authentication and enable a new verification of cough-based user authentication as it includes different modalities from a large pool of users.
- **Demographic prediction**: Our dataset provides an opportunity of exploring machine learning for cough/breathing based demographic perception.
- **Respiratory disease detection**: Our collected dataset contains cough/breathing sounds along with medical history at a larger scale, and this could facilitate research in developing machine learning models based on different types of sounds.
- **Smoking status detection**: Exposure to tobacco smoke may lead to a variation in speech sounds, and our data can accelerate such research.
- **Semi-supervised and self-supervised learning**: For audio-based health applications with the large number of unlabelled, it is a natural next step to explore semi-supervised methods to boost the performance. Our data can be used for model development and evaluation.

References

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